

Spectral Gradient Filtering on Numerai

What we tried, what went wrong, what eventually worked, and what remains unknown



Audited bounded result

Spectral pipeline: **+0.00307
CORR**

3/3 random seeds win

CORR Sharpe: **1.557 vs 1.265**

This is evidence to continue, not yet a leaderboard claim.

Executive answer

What finished

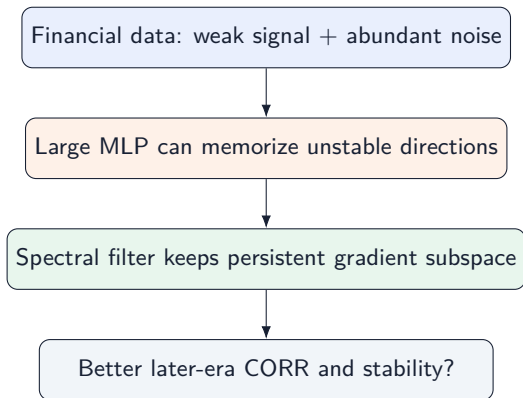
- ▶ Corrected parameter-space spectral implementation
- ▶ Low- and high-rank exploratory studies
- ▶ Overparameterized 13.7M-parameter stress test
- ▶ Symmetric development selection on Numerai v5.3
- ▶ Frozen, chronological outer test and six-cell audit

What it means

- ▶ **Serious competitor: yes**
- ▶ Spectral pipeline beat selected AdamW by 15.5% relative CORR
- ▶ High-rank filtering strongly suppressed memorization
- ▶ **Not an optimizer-only causal comparison**
- ▶ **Not yet leaderboard-comparable or submission-ready**

The next experiment should isolate the optimizer while extending the untouched time horizon.

The motivating hypothesis

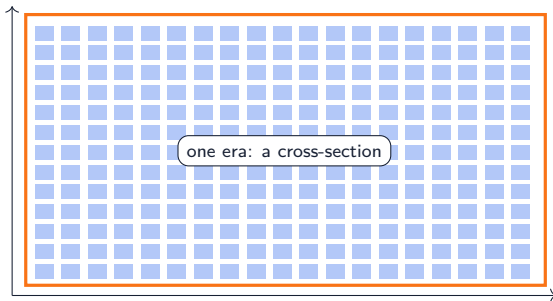


Why Numerai?

- ▶ Extremely low signal-to-noise ratio
- ▶ Many anonymized candidate predictors
- ▶ Time and regime shift matter
- ▶ Exact per-era evaluation
- ▶ Public competitive benchmark

What the dataset looks like

stocks / rows



3,555 anonymized numeric features used here

Rows, eras, target

- ▶ Each row represents one anonymized security at one historical date
- ▶ An **era** groups rows from the same market period
- ▶ Features are anonymized engineered signals; semantics are intentionally hidden
- ▶ Primary label: Numerai v5.3 main target
- ▶ Model output is ranked within each era before CORR scoring

The task is cross-sectional prediction through time—not forecasting one price series.

Train / validation / test through time

Numerai evaluation respects time: whole eras, purge, then later eras

Hyperparameters selected only inside the blue region

Three frozen seeds; no reselection



Train

Fit weights on earlier eras.

Inner validation

Choose architecture and optimizer settings inside training history.

Outer test

Evaluate one frozen decision on immediately later eras.

In production one would periodically retrain on newly resolved labels. The experiment instead freezes one historical decision so its out-of-time evidence remains interpretable.

What exactly is being optimized?

AdamW baseline

$$\theta_{t+1} = \theta_t - \eta \text{AdamW}(g_t)$$

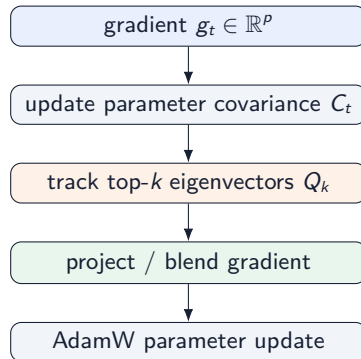
Uses the full minibatch gradient, adaptive coordinate scaling and decoupled weight decay.

Spectral wrapper

$$C_t = \beta C_{t-1} + (1 - \beta) g_t g_t^\top, \quad C_t \in \mathbb{R}^{p \times p}$$

$$C_t Q_k = Q_k \Lambda_k, \quad \tilde{g}_t = (1 - \alpha) g_t + \alpha Q_k Q_k^\top g_t$$

AdamW receives the filtered gradient $\tilde{g}_t$.



The first implementation question mattered

Wrong object for the intended optimizer

A $B \times B$ batch Gram matrix describes relationships among the B examples currently sampled. Its eigenvectors live in *sample space*.

$$GG^T \in \mathbb{R}^{B \times B}$$

**not the persistent parameter directions
requested**

Correct object tested later

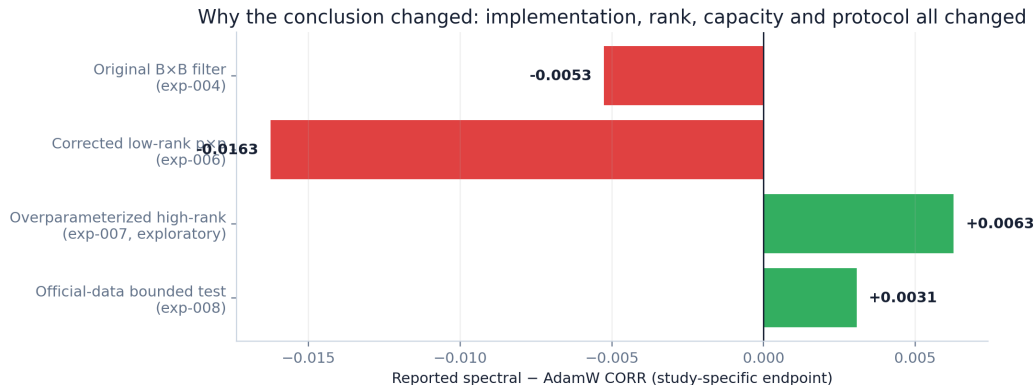
A $p \times p$ covariance (implemented with a low-rank basis) tracks directions in the model's parameter space across updates.

$$G^T G \in \mathbb{R}^{p \times p}$$

eigenvectors are parameter directions

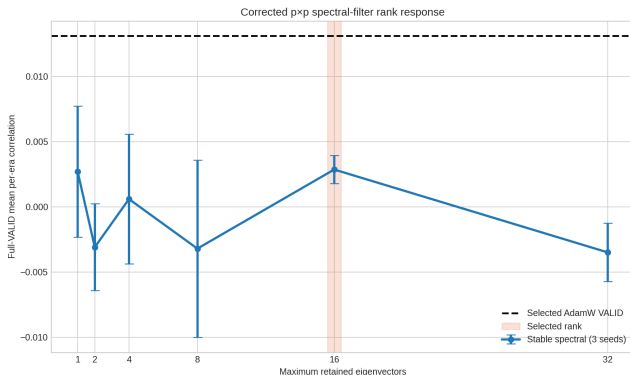
The original negative result and the corrected high-rank result are studies of materially different interventions.

The complete research arc



Effect sizes are shown to explain direction and chronology; they are not pooled because datasets, capacities, endpoints and protocols differed.

Phase I: corrected low rank still failed

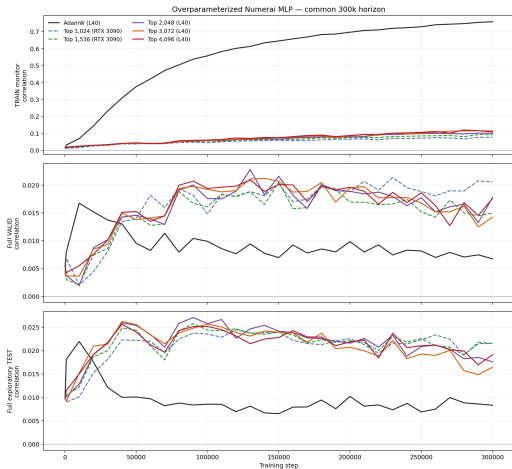


Result

- ▶ Corrected $p \times p$ hard filter
- ▶ Ranks 1–32, three selection replicas
- ▶ Rank 16 selected on validation
- ▶ Five-seed untouched test
- ▶ AdamW: 0.01686 CORR
- ▶ Spectral: 0.00061 CORR
- ▶ $\Delta = -0.01625$

Lesson: correctness alone was insufficient. A tiny rank can discard almost the entire useful update in a multi-million-dimensional model.

Phase II: force the model to overfit



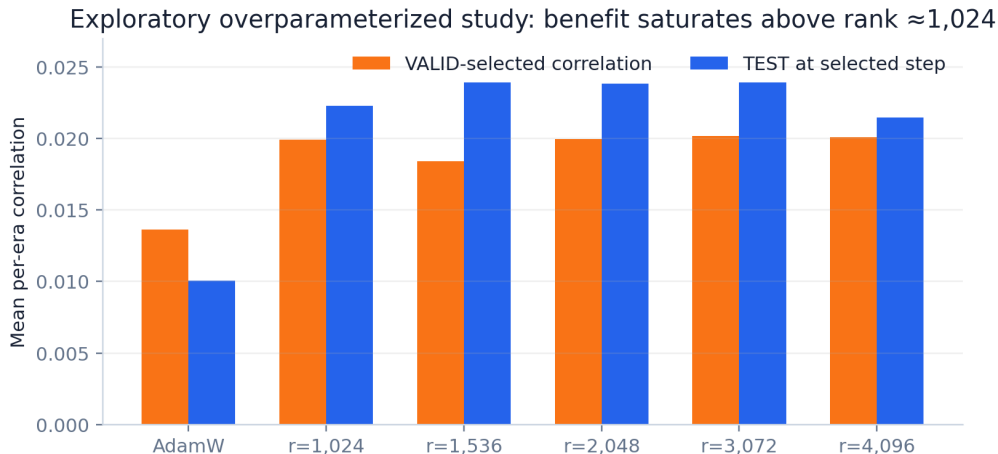
Stress test

- ▶ 13.7M-parameter MLP
- ▶ Same architecture and batch stream
- ▶ AdamW vs top- k spectral projection
- ▶ $k = 1,024$ to 4,096
- ▶ Up to one million updates

Observed mechanism

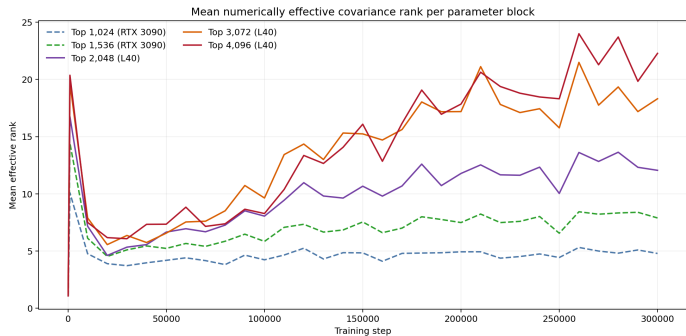
AdamW reached 0.76 train CORR while validation deteriorated. Spectral stayed near 0.07–0.11 and retained later-era signal.

High rank changed the answer



Exploratory selection used validation trajectories; the displayed TEST trajectory was repeatedly observed and is not an untouched final holdout. Ranks 1,024/1,536 ran on RTX 3090; others on L40.

Requested rank is not realized rank



Key diagnostic

Requested global ranks of 1,024–4,096 produced mean effective ranks of only roughly 5–22 after blockwise eigenspectrum thresholding.

Interpretation

Benefits saturated because the useful persistent subspace was much smaller than the nominal capacity. Rank is a ceiling, not necessarily the number of active directions.

Why we did not run the six-week exhaustive plan

Original ambition

- ▶ 40 paired configurations
- ▶ Multiple update budgets
- ▶ Multiple inner folds and seeds
- ▶ High-rank amendments
- ▶ Three nested outer blocks
- ▶ Sealed validation and live packaging

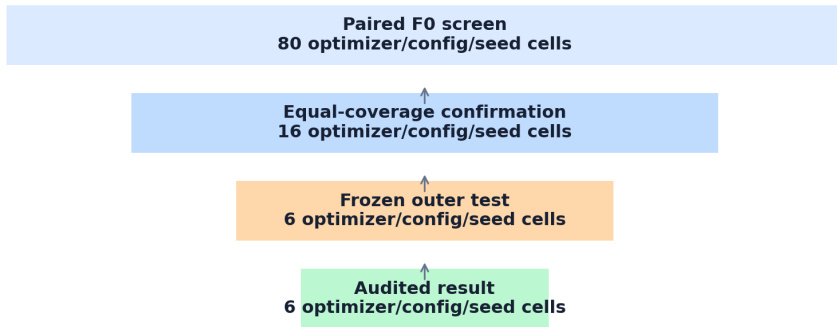
Bounded decision

- ▶ User requested a one-day answer
- ▶ Runtime cannot select winners
- ▶ Incomplete F1 results excluded
- ▶ Complete F0 nominates two IDs
- ▶ Equal confirmation coverage
- ▶ One untouched outer block

Goal: decide whether spectral is credible enough to continue—not certify a leaderboard submission.

The bounded protocol

Bounded one-day protocol: selection narrows before untouched outcomes

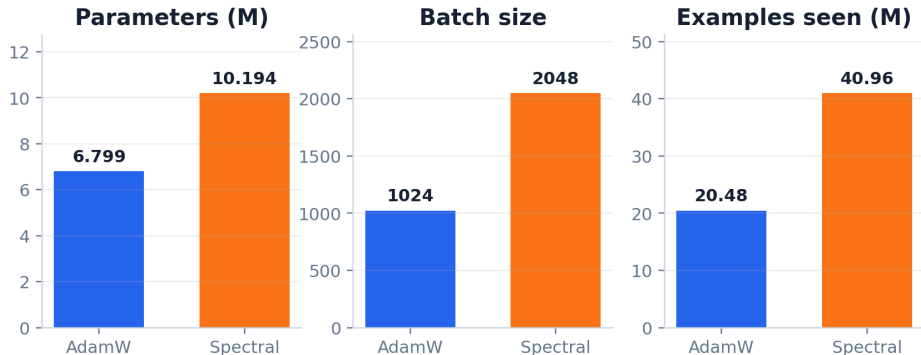


Leakage controls

Candidate nomination used only the complete 5,000-update screen. Both optimizers then ran both IDs 38 and 39 on two inner folds and two seeds. One winner per optimizer was frozen before any outer score existed.

Why AdamW had 6.8M parameters and spectral had 10.2M

The outer comparison estimates best selected pipelines—not an optimizer-only effect



Why it happened: architecture was treated as a hyperparameter. Each optimizer independently selected the pipeline with highest development CORR from the same paired union.

What it answers: “Which selected training pipeline is better?” It does **not** answer “Does spectral filtering improve this exact MLP?”

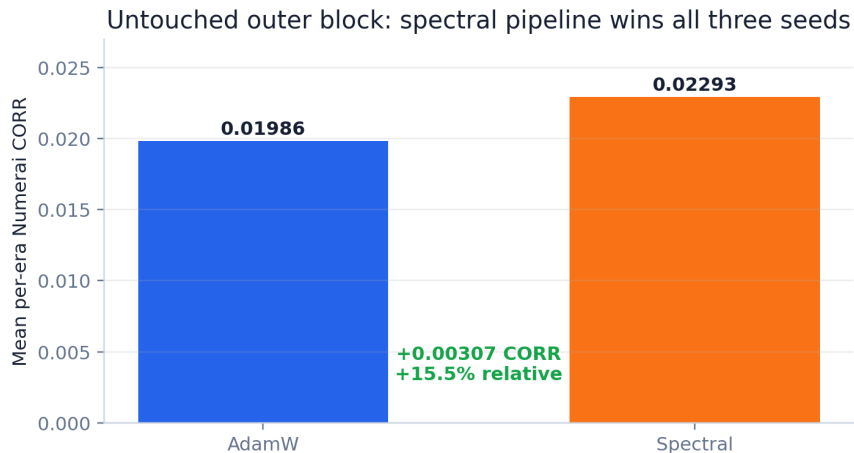
The selected pipelines were materially different

	AdamW winner: config 38	Spectral winner: config 39
Model	4 layers, width 1,024, ReLU	3 layers, width 1,536, SiLU
Parameters	6,799,361	10,194,433
Features	all 3,555	all 3,555
Batch	1,024, era-batched	2,048, row-batched
Examples	20.48M	40.96M
Learning rate	1.2876×10^{-4} , cosine	2.5188×10^{-3} , constant
Regularization	dropout 0.10, small weight decay	dropout 0.02, no weight decay, clip 5
Spectral filter	none	learned, rank 128, strength 1, decay 0.99

Fairness verdict

Symmetric as a *best-pipeline competition*; asymmetric as a *mechanistic optimizer ablation*. The next experiment must run both optimizers on both frozen architectures in the untouched period.

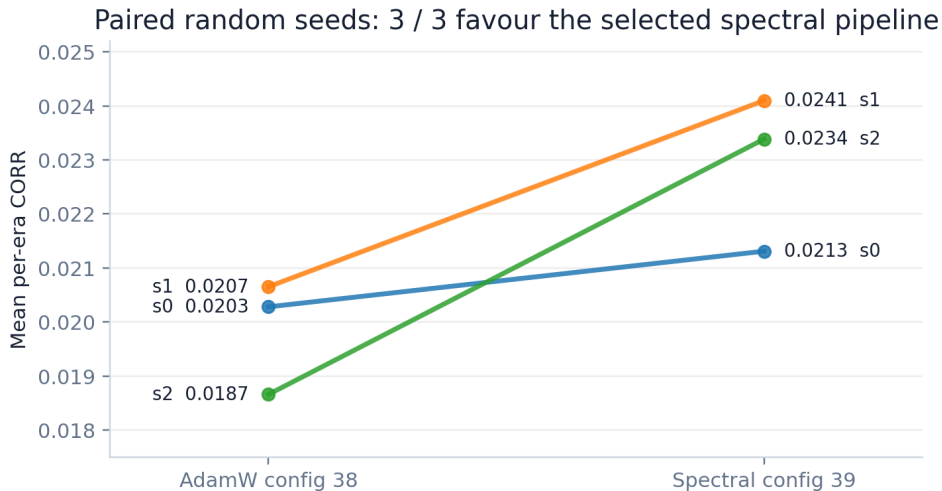
Audited outer result



359,860 rows per seed; frozen configurations; three seeds; exact standalone Numerai CORR; artifact hashes and metric reproduction audited.

78 later eras ×

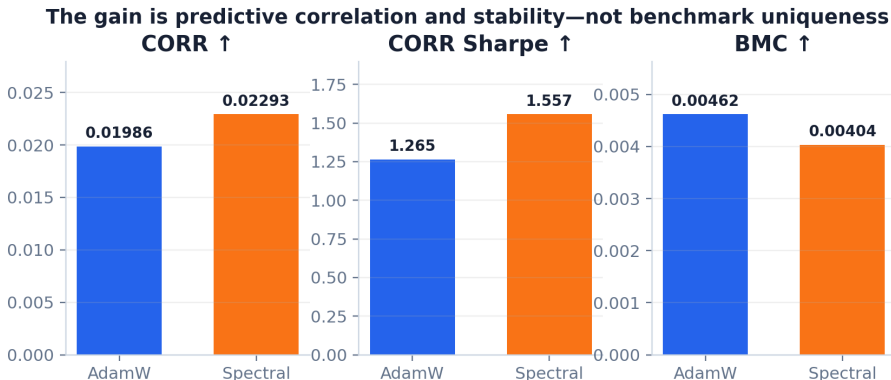
Every random seed favoured spectral



differences: +0.00103, +0.00345, +0.00473. Mean: +0.00307 CORR.

Seed-paired

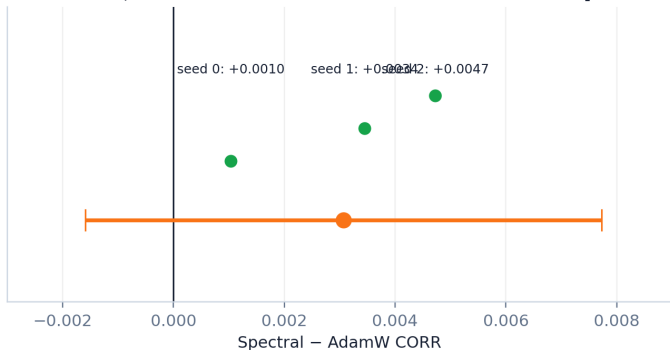
The tradeoff: stronger signal, slightly less uniqueness



- ▶ CORR improved by 15.5% relative; CORR Sharpe improved by 23.1%.
- ▶ BMC fell by 0.00058: slightly less contribution beyond Ender60.
- ▶ Acceptable for standalone prediction, but relevant to Numerai payout and model portfolios.

How certain are we?

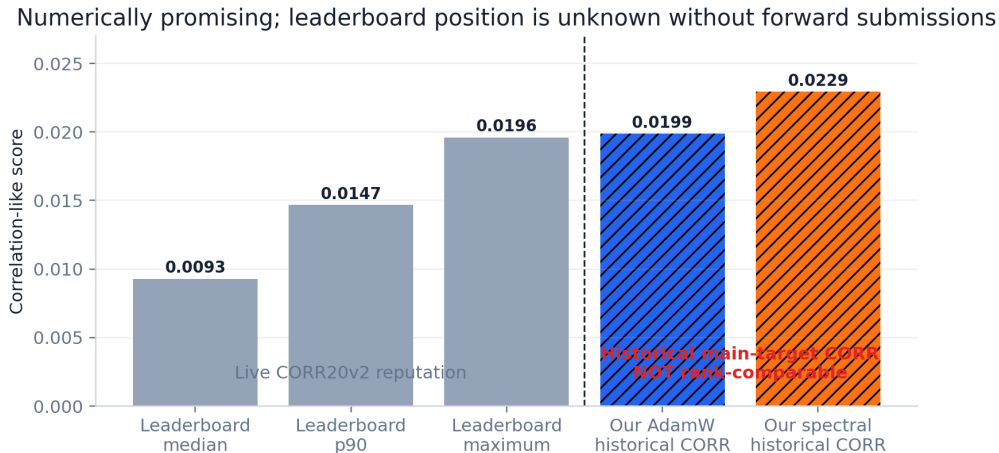
Three consistent wins, but seed-level 95% t interval crosses zero $[-0.0016, +0.0077]$



Honest reading

Three out of three wins are encouraging. But a seed-level t interval is wide with $n = 3$ and crosses zero. An era-block bootstrap and two additional untouched outer blocks are needed for a strong statistical claim.

Where are we versus the Numerai leaderboard?



Snapshot: round 1330, 12 Aug 2026, top 1,000 models. Live leaderboard CORR20v2 reputation and our historical v5.3 main-target CORR use different targets, horizons and sampling. The numbers can motivate a submission; they cannot imply a rank.

What competitors get—and why comparison is hard

Public metric	median	90th pct.	maximum / caveat
CORR20v2 reputation	0.00926	0.01466	0.01959; live rolling reputation
CORJ60 reputation	0.01054	0.01918	0.03262; different neutralized target
BMC reputation	0.00633	0.00967	0.01720; contribution relative to benchmark ensemble
MMC reputation	0.00399	0.00658	0.01262; contribution relative to metamodel
Our spectral outer CORR	0.02293		historical main-target result; not live reputation
Our spectral BMC	0.00404		historical Ender60-relative contribution

Only decisive comparison

Freeze the procedure, create a resource-valid live callable, submit unstaked for enough rounds, then compare like-for-like live reputations.

What the result does—and does not—establish

Supported

- ▶ Spectral is worth continued investigation
- ▶ Selected spectral pipeline generalized better on outer block 1
- ▶ High-rank filtering can arrest MLP memorization
- ▶ Benefits are rank- and regime-dependent
- ▶ Runtime disadvantage is irrelevant to this scientific question

Not supported yet

- ▶ Spectral filtering alone caused $+0.00307$
- ▶ Rank 128 is globally optimal
- ▶ The gain persists across all historical regimes
- ▶ The model beats current live competitors
- ▶ A production submission will have positive payout metrics

The highest-value next experiment

Freeze architectures 38 and 39; run AdamW and spectral on **both**



Use ranks 128, 512, 1,024, 2,048 (memory-probed); same examples and batch streams



Evaluate frozen choices on outer blocks 2 and 3; era-block bootstrap



If positive: sealed official validation, production refit and unstaked live candidate



Only live rounds answer the leaderboard question

Go / no-go criterion

Continue only if spectral improves exact CORR on both architectures or preserves CORR while materially improving stability/BMC across outer blocks.

Future work, ordered by information per GPU-hour

Priority	Experiment	Decision unlocked
1	Matched-architecture 2×2	Separates optimizer effect from model/HPO effect
2	Outer blocks 2 and 3	Tests regime robustness and tightens uncertainty
3	Rank ladder to 2,048+	Locates saturation under the official-data pipeline
4	Equal examples / compute diagnostics	Separates more data exposure from filtering
5	Era-block bootstrap	Quantifies temporal rather than seed-only uncertainty
6	Sealed official validation	Produces historical score with frozen procedure
7	Unstaked live submission	Establishes actual leaderboard comparability
8	Continual retraining simulation	Approximates the real finance deployment loop

Bottom line



The spectral idea survived the go/no-go test.

It won all three frozen outer seeds and improved both average CORR and stability.

But the selected pipelines differed in architecture, batch size and examples seen.

The result justifies one controlled follow-up—not a leaderboard claim.

Appendix: why the original theoretical story weakened

For scalar-output losses, a per-example gradient often factors as

$$g_i = \ell'_i(f_\theta(x_i), y_i) \nabla_\theta f_\theta(x_i).$$

Normalizing g_i can remove the residual magnitude and leave mainly a sign times the model Jacobian direction. In the original $B \times B$ construction, target permutation therefore changes signs but can preserve the eigenspectrum. “Gradient consensus” may measure shared Jacobian / feature structure rather than target signal.

Why the current result can still be real

The corrected learned $p \times p$ filter acts as a history-dependent optimizer regularizer. It may suppress rapidly changing parameter directions even if the original label-consensus interpretation is incomplete.

Appendix: audit trail

Artifact	Role
selection-one-day-f0-top1.json	Frozen F0 nominations from complete equal-coverage screen
selection-one-day-confirmed-top1.json	Development-only winners and score-file hashes
submission-one-day-outer1.tsv	Exact six-cell frozen outer manifest
outer-results.csv	Six metrics rows with signatures and artifact hashes
outer-audit.json	Split, seeds, selected IDs, rows and manifest hash
commit 7dac4cd	Allows sub-ppm float32 serialization roundoff while retaining metric-reproduction checks

The audit reproduced metrics from stored predictions, verified all 78 expected outer eras, checked row uniqueness/alignment, configuration provenance, model signatures, and prediction/result SHA-256 hashes.

Appendix: evidence map

Phase	Question	Outcome	Main limitation
exp-001–005	Does original $B \times B$ filter transfer?	Negative; target-independence insight	Different intervention from intended $p \times p$ optimizer
exp-006	Does corrected low-rank hard projection work?	No; rank 16 severely harmed CORR	Tiny rank, small model, limited tuning
exp-007	Can high rank prevent memorization?	Yes, strongly in overparameterized MLP	Exploratory TEST trajectory; one seed
exp-008	Is selected spectral pipeline a serious competitor?	Yes: +0.00307 outer CORR, 3/3 seeds	Different selected architectures; one outer block

Questions?

Recommended next step: matched-architecture outer evaluation before sealed validation.